Marked Polygon Correlations and One-Arc Bounds

Uniform Cauchy-Trace Remainders and Stable Denominator Certificates

Dr. Muralidhara Subbarao
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Stony Brook University
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Abstract

We analyze the finite interpolation and denominator mechanism in the September 26, 2026 manuscript *Marked polygon correlations and one-arc bounds*. Its positive Pfaffian representation and exact Cauchy identity are retained, not claimed as new. An absolute-integral localization argument sharpens its uniform logarithmic denominator estimate to $\mathscr{D}_N \geq e^{-C} N^{5/24}$, including coincident nodes and arbitrarily unequal two-height multiplicities. The associated Jensen trace certificate, rather than the full correlation, is Lipschitz in the mean displacement of the imaginary nodes, with a constant independent of their number. We give explicit analytic error budgets and a finite integration truncation bound that avoids an inverse at confluent nodes. The physical spin interpretation, magnetic numerator transport, one-arc integration and planar geometric transfer remain separate source obligations. Original Ram localization, absolute-moment and finite-coordinate proofs inform the error organization; classical Cauchy, Pfaffian and Fourier identities retain their usual attribution. Contribution assessment: C3/5, a uniform bounded-error denominator refinement and quantitative node-stability certificate are proved without upgrading the imported polygon or one-arc theorems.

Contents

1 Model, normalization and proof boundaries **2**

2 Exact finite reconstruction and its limits **3**
2.1 Where positivity actually enters 3

3 Fourier trace in the physical displacement variable **4**

4 A bounded-error denominator theorem **4**

5 Node-height stability without inverse conditioning **6**
5.1 Examples and distinctions 7

6 A finite integration certificate **7**

7 Unequal arcs and the global dependency map **8**
7.1 What the local gain transfers 8
7.2 The source's physical and analytic route 8

8 Original Ram comparison and current literature **9**

9 Advantages, costs and further work **10**

10 Verification and contribution boundary **10**

1 Model, normalization and proof boundaries

The source [1] uses the honeycomb lattice dual to a triangular lattice of side one. A polygon is simple, unoriented and unrooted; its length counts visited vertices. Its critical physical weight is $\rho_{\mathrm{v}}^{|P|}$, where

$$\rho_{\mathrm{v}} = (2+\sqrt{2})^{-1/2}, \qquad \lambda = \pi/8, \quad \beta = \pi/4, \quad \gamma = 4/3, \quad d = \cos(\pi/4).$$

Translation classes are taken with respect to the triangular translation lattice. A marked port is an edge midpoint; its half-edge convention must be retained when counting visited vertices.

A cyclic row has order $[p, X, p', Y]$, with $s = |X|$, $b = |Y|$ and $N = s + b + 2$ even. All real spectral offsets vanish in the physical application. The Y sites have height zero; the X sites and marks have heights in $\{0, \beta\}$. The logarithmic wall lengths are

$$L_X = \gamma^{-1} \log s, \qquad L_Y = \gamma^{-1} \log b.$$

They are not geometric arc lengths. The fraction t of upper-height ordinary X sites is another parameter, not a particle coordinate.

The source claims a sharp marked-pair estimate

$$\log \mathcal{C}(p, p') \le -\frac{4}{3} \log s + \frac{617}{1200} \log(b/s) + o(\log b) \tag{1}$$

uniformly when $c_1 L_Y \le L_X \le L_Y$, $t \ge 1/2$, and, if $t < 1$, $s(1-t) \ge e^{c_4 L_Y}$, with fixed admissible $c_1, c_4 > 0$. It separately claims a coarse bound $A(p, p') \le b^C$ for a single west-to-east arc, assuming $L_Y \ge L_X \ge L_*$ and $t \ge 1/2 - \delta_1$. That result has no lower bound on L_X/L_Y. Neither statement follows just by taking a matrix inverse.

Here the new local object is the cleared vacuum denominator. Write

$$H_*(u) = \frac{e^{2u} + 2de^u + 1}{(e^u - 1)e^{u/2}} \tanh(\gamma u/2), \tag{2}$$

$$\mathscr{D}_N(a) = \left| \frac{\widetilde{h}_N(a)^2}{(\gamma/2)^N \prod_{i<j} H_*(a_i - a_j)^2} \right|. \tag{3}$$

The vacuum polynomial $\widetilde{h}_N$ uses the source's specified polynomial scaling. The associated physical half-cylinder state $U_N = V_N/h_N$, not this polynomial, has empty matching component one. For distinct nodes, an equivalent definition, independent of any physical probability interpretation, is

$$\mathscr{D}_N = (\gamma/2)^{-N} \left| \frac{\mathrm{Pf}[f(a_i - a_j)]}{\mathrm{Pf}[f_0(a_i - a_j)]} \right|^2, \quad f(u) = \frac{\sinh u}{\cosh u + d}, \quad f_0(u) = \tanh(\gamma u/2). \tag{4}$$

We use the continuous cleared extension at collisions. A common imaginary shift changes none of these difference kernels. Thus the two-height problem lies in the larger domain

$$a_i = i y_i, \qquad -\beta/2 \le y_i \le \beta/2, \qquad N \ge 2 \text{ even}. \tag{5}$$

This continuous-node domain is analytic; intermediate heights are not asserted to be physical honeycomb weights.

2 Exact finite reconstruction and its limits

Let $K(u) = \operatorname{sech}(\gamma u/2)$, $K[a]_{ij} = K(a_i - a_j)$, and set

$$B_a(z) = \prod_{i=1}^{N} \frac{e^{\gamma z} - e^{\gamma a_i}}{e^{\gamma z} + e^{\gamma a_i}}.$$

For distinct nodes in (5), $K[a]$ is invertible. Indeed, with $t_i = e^{\gamma a_i}$, its entries are $2t_i^{1/2}t_j^{1/2}/(t_i + t_j)$, using consistent square roots. The Cauchy determinant is

$$\det\left[\frac{1}{t_i + t_j}\right]_{i,j=1}^{N} = \frac{\prod_{i<j}(t_j - t_i)^2}{\prod_{i,j}(t_i + t_j)}.$$

Distinctness and $|\arg t_i| \le \pi/6$ make every displayed factor nonzero. This is a bilinear Cauchy matrix, not a Hermitian Gram matrix.

Lemma 2.1 (Classical Cauchy reconstruction)**.** *At regular z, w,*

$$K[z, a]K[a]^{-1}K[a, w] = K(z - w)\{1 - B_a(z)B_a(w)\}. \tag{6}$$

Proof. Apply the rectangular Cauchy determinant formula to the bordered matrix with row parameters $(t_1, \dots, t_N, e^{\gamma z})$ and column parameters $(t_1, \dots, t_N, e^{\gamma w})$, and divide by its unbordered determinant. The Schur complement is the interpolation error. Cancellation of the Cauchy products gives $B_a(z)B_a(w)$ times $1/(e^{\gamma z} + e^{\gamma w})$. Restoring the row and column square-root factors proves the identity. Analytic continuation extends it away from the initial generic parameters. □

This identity is already in the source and is classical [2]. At repeated nodes $K[a]$ is singular. Equation (6)'s right side, not a nonexistent inverse, defines its confluent interpolation expression. No condition-number bound for the raw matrix is implied. Similarly, the source's occupied vacuum components have square-root factors and degree at most $N - 2$ in the relevant variable; the vacant components have degree at most $N - 1$. Fusion zeros determine those particular polynomials only after this degree bound and a nonzero normalization are supplied. A generic finite interpolation slogan would omit both requirements.

2.1 Where positivity actually enters

For $N = 2m$, the source's integral formula for f_0 uses

$$f_0(u) = \int_0^\infty (e^{iku} - e^{-iku}) \frac{dk}{i\gamma \sinh(3\beta k)}.$$

The formula for f multiplies this measure by $(\gamma/2)(1 + (2\cosh(2\beta k))^{-1})$. Expanding the exterior power gives a determinant with columns $(e^{-k_j y_i}, e^{k_j y_i})$ on $0 < k_1 < \cdots < k_m$. Its sign is fixed after ordering the distinct y_i's: a nonzero linear combination of r real exponentials with distinct exponents has at most $r - 1$ zeros, by division by the first exponential and Rolle induction. Column-pair subtraction removes the apparent singularity at $k_j = 0$; the strict imaginary strip controls infinity. This proves the sign assertion needed to normalize a reference probability $\mathbb{P}_0$. It does not make the magnetic gas positive.

Put

$$S(k) = \log\left(1 + \frac{1}{2\cosh(2\beta k)}\right). \tag{7}$$

The source's exterior-power calculation, including the square and $(\gamma/2)^N$, now gives

$$\mathscr{D}_N = \left(\mathbb{E}_0 e^{\sum_{j=1}^m S(k_j)}\right)^2, \qquad \log \mathscr{D}_N \geq 2\mathbb{E}_0 \sum_{j=1}^m S(k_j) =: \mathscr{J}_N(a). \tag{8}$$

These positive-Pfaffian and Jensen steps are retained source results, not the increment of this report.

3 Fourier trace in the physical displacement variable

Our Fourier convention is $\widehat{u}(k) = \int_{\mathbb{R}} e^{-ikr} u(r)\, dr$, with inverse factor $1/(2\pi)$. Define

$$\widehat{K}(k) = \frac{2\pi}{\gamma}\operatorname{sech}(3\beta k), \quad T(k) = S(k - i\gamma/2), \quad \widehat{J}(k) = \widehat{K}(k)T(k).$$

The logarithm in S is continued from its real value on the real axis within $|\Im k| < \pi/(4\beta)$. Both its argument's poles and zeros lie outside this open strip. Since $\gamma/2 = \pi/(6\beta)$, the shift defining T is legitimate.

Fix once and for all

$$\beta/2 < h < \beta, \qquad \widehat{g}(-k) = \frac{T(k)}{2\widehat{K}(k)\cosh(2hk)}, \quad W_\pm(r) = g(r)K(r \pm 2ih). \tag{9}$$

The symbol on the right and each fixed derivative decay exponentially: the net exponential factor is $e^{-(2h-\beta)|k|}$. Hence that symbol is Schwartz, as is g. This argument uses smooth derivative bounds on the real line; it does not need a claimed exponentially decaying inverse Fourier transform. Also $K(r \pm 2ih)$ has no real pole and decays at infinity. In particular

$$\kappa_j = \sum_{\pm} \int_{\mathbb{R}} |r|^j |W_\pm(r)|\, dr < \infty \quad (j = 0, 1). \tag{10}$$

For distinct nodes, differentiation of the deformed Pfaffian with spectral factor $1 + tS$ yields $\mathscr{J}_N = \operatorname{Tr}(K[a]^{-1}J[a])$. The constant added to the odd kernel in this conversion is $1 + tS(0)$, not just one: adding an all-ones rank-one matrix to an even skew matrix preserves its determinant since $\mathbf{1}^{\mathrm{t}}A^{-1}\mathbf{1} = 0$. The exponential conjugation shifts the Fourier multiplier to T. Combining the finite trace with (6) gives

$$\mathscr{J}_N(a) = \sum_{\pm} \int_{\mathbb{R}} W_\pm(r) I_\pm(r; a)\, dr, \quad I_\pm = \int_{\mathbb{R}} [1 - B_a(x + r/2 \pm ih) B_a(x - r/2 \mp ih)]\, dx. \tag{11}$$

For completeness, integrating the two K factors first in x gives multiplier $\widehat{K}(k)^2 e^{\mp 2hk}$. The sign sum and the r integration in (9) leave $\widehat{K}T$, exactly the kernel J. The bounds below justify absolute integration for each fixed N and extend (11) continuously to repeated nodes.

The form (11) retains the physical displacement r, its Lebesgue measure and the full complex weight $W_\pm$. It is an exact localization of the finite trace, not a positive quadrature rule or a reconstruction of arbitrary contour functions.

4 A bounded-error denominator theorem

Define constants depending only on the fixed contour height:

$$c_0 = \cos(\gamma(h + \beta/2)) > 0, \quad c = c_0/2, \quad D_h = \frac{2}{\gamma}\left(\frac{e^{-c}}{c} + 2\right), \quad C_h = 2D_h\kappa_0 + \kappa_1. \tag{12}$$

These are explicit analytic budgets, not numerically certified values. In particular, κ_j still require evaluation with error bounds if a numerical prefactor is wanted.

Lemma 4.1 (Uniform product estimates)**.** *For either sign and every node list* (5)*,*

$$|B_a(x \pm ih)| \le e^{-cNe^{-\gamma|x|}}, \tag{13}$$

$$|1 - B_a(x \pm ih)| \le 2Ne^{-\gamma|x|}. \tag{14}$$

Proof. Write $R = e^{\gamma x}$ and $\theta_i = \gamma(\pm h - y_i)$. One factor b_i has

$$|b_i|^2 = \frac{R^2 + 1 - 2R\cos\theta_i}{R^2 + 1 + 2R\cos\theta_i}, \qquad \cos\theta_i \ge c_0.$$

Since $R/(R+1)^2 \ge \min(R, R^{-1})/4$, $1 - |b_i|^2 \ge c_0 \min(R, R^{-1})$. Using $\log(1-u) \le -u$ proves (13), including any zero factor by continuity. In particular every factor has modulus at most one. For $x \ge 0$, $|1 - b_i| \le 2/R$; for $x \le 0$, $|1 + b_i| \le 2R$, because the denominator modulus is at least $\max(1, R)$. Telescope products against the factors 1, or against -1, respectively. At the negative end $\prod_i(-1) = 1$ uses that N is even. This proves (14). □

Lemma 4.2 (Uniform transition mass)**.** *Put $L = \gamma^{-1}\log N$ and $b_L(x) = \mathbf{1}_{\{|x|>L\}}$. Then*

$$\|B_a(\cdot \pm ih) - b_L\|_{L^1(\mathbb{R})} \le D_h. \tag{15}$$

Moreover

$$|I_\pm(r; a) - 2L - \min(|r|, 2L)| \le 2D_h \quad (r \in \mathbb{R}). \tag{16}$$

Proof. Inside the window, (13) and $u = Ne^{-\gamma x}$ on its positive half give

$$\int_{-L}^{L} |B_a(x \pm ih)|\, dx \le \frac{2}{\gamma}\int_1^N \frac{e^{-cu}}{u}\, du \le \frac{2e^{-c}}{\gamma c}.$$

Outside, (14) integrates to at most $4/\gamma$. This proves (15). For factors of modulus at most one, $|AB - ab| \le |A - a| + |B - b|$. Apply this with the shifted B's and shifted step functions; translation invariance preserves their L^1 errors. The integral of $1 - b_L(x + r/2)b_L(x - r/2)$ is the length of the union of two intervals of length $2L$ whose centers are $|r|$ apart. That length is $2L + \min(|r|, 2L)$. This proves (16) even for $|r| > 2L$, without a shared central interval or a positivity assumption on the complex integrand. □

Theorem 4.3 (Uniform bounded trace remainder)**.** *For all even $N \ge 2$ and all lists* (5)*, with the continuous trace definition* (11) *at collisions,*

$$\left|\mathscr{J}_N(a) - \frac{5}{24}\log N\right| \le C_h, \qquad \mathscr{D}_N(a) \ge e^{-C_h}N^{5/24}. \tag{17}$$

The constants do not depend on N, the number of distinct heights, or their multiplicities.

Proof. From (16),

$$\left|\mathscr{J}_N - 2L\sum_{\pm}\int_{\mathbb{R}} W_\pm(r)\, dr\right| \le 2D_h\kappa_0 + \kappa_1.$$

Fourier inversion gives $\sum_\pm \int W_\pm = \int T(k)\, dk/(2\pi)$. Move this last contour to the real axis inside the stated holomorphic strip, where S decays exponentially. For $k > 0$, put $v = e^{-2\beta k}$; then

$$S(k) = \log(1 - v^3) - \log(1 - v) - \log(1 + v^2).$$

Integration of each absolutely integrable logarithmic series yields

$$\int_{\mathbb{R}} S(k)\,dk = \frac{1}{\beta}\left(-\frac{\pi^2}{18}+\frac{\pi^2}{6}-\frac{\pi^2}{24}\right) = \frac{5\pi^2}{72\beta}.$$

Therefore $2L\int T/(2\pi) = 5\log N/24$. For distinct nodes, (8) gives the lower bound for $\mathscr{D}_N$. For coincident nodes, the integrable bounds in Lemmas 4.1–4.2 imply continuity of the trace at fixed N. The scalar Pfaffian ratio is taken after clearing its Vandermonde factors. Equivalently use the source's polynomial vacuum and note

$$H_*(iy) = \frac{(\cos y + d)\tan(\gamma y/2)}{\sin(y/2)}, \quad |y| \le \beta, \qquad H_*(0) = \gamma(1+d) > 0.$$

This is finite and positive throughout the allowed interval. Thus the cleared ratio is continuous, and the same lower bound passes to the limit. No inverse of the collided matrix is taken. □

Remark 4.4 (What was strengthened). The source's Theorem `thm:06-denominator` already proves a uniform lower bound $N^{5/24-o(1)}$. Its proof already contains (13)–(14) and the exact trace formula. The increment is their uniform L^1 transition control, yielding a bounded logarithmic loss and hence a fixed prefactor. This is neither a new Cauchy identity nor a two-sided asymptotic $\mathscr{D}_N \sim cN^{5/24}$. Jensen's unestimated excess can remain large.

5 Node-height stability without inverse conditioning

Theorem 5.1 (Average-displacement certificate)**.** *For two lists $a_i = iy_i$, $a'_i = iy'_i$ satisfying* (5) *with the same even N,*

$$|\mathscr{J}_N(a) - \mathscr{J}_N(a')| \le \frac{16\kappa_0}{cN}\sum_{i=1}^{N}|y_i - y'_i|. \tag{18}$$

Because the products are symmetric in their nodes, one may minimize over matchings on the right. It equals $(16\kappa_0/c)W_1(\mu_a,\mu_{a'})$, where $\mu_a = N^{-1}\sum_i \delta_{y_i}$ and W_1 is the usual one-dimensional transport distance.

Proof. Interpolate linearly between matched node heights, staying in the closed interval. For $u = e^{\gamma(x\pm ih)}$, $t = e^{i\gamma y_i}$, the derivative of the i-th factor is

$$\partial_{y_i}\frac{u-t}{u+t} = -\frac{2i\gamma ut}{(u+t)^2}, \qquad |\partial_{y_i} b_i| \le 2\gamma e^{-\gamma|x|}.$$

Multiplying the other $N-1$ factors and applying (13) gives

$$|\partial_{y_i} B_a(x \pm ih)| \le 2\gamma e^{-\gamma|x|}e^{-c(N-1)e^{-\gamma|x|}}.$$

Its integral is at most $4/[c(N-1)] \le 8/(cN)$. Integration along the interpolation path gives the L^1 bound

$$\|B_a(\cdot \pm ih) - B_{a'}(\cdot \pm ih)\|_1 \le \frac{8}{cN}\sum_i |y_i - y'_i|.$$

The difference of the two shifted products in (11) has twice this bound, independently of r. Integrate against $|W_\pm|$ and sum signs. Collisions cause no problem for this product proof. Sorted matching minimizes the absolute displacement: uncrossing two crossed pairs cannot increase their total distance. This proves the stated W_1 interpretation. □

5.1 Examples and distinctions

For $N = 2$, with height separation $v = |y_1 - y_2| \le \beta$, the exact cleared ratio is

$$\mathscr{D}_2(v) = \left[\frac{2\sin v}{\gamma(\cos v + d)\tan(\gamma v/2)}\right]^2, \qquad \mathscr{D}_2(0) = \left[\frac{4}{\gamma^2(1+d)}\right]^2. \tag{19}$$

This concrete confluence limit is positive while the raw 2-by-2 matrix $K[a]$ loses rank. In the two-height family, moving ℓ sites from one level to the other changes the trace certificate by at most $16\kappa_0\beta\ell/(cN)$. Thus a bounded number of changed sites costs $O(N^{-1})$, even at extreme multiplicities. For a height grid of mesh Δ, rounding each node to an allowed nearest grid value costs at most $8\kappa_0\Delta/c$. One must keep N nodes, including multiplicities; replacing them by only the distinct grid sites changes the problem.

This is stability of a lower-bound certificate. It is not a bound on $|\log \mathscr{D}_N(a) - \log \mathscr{D}_N(a')|$, because the difference between $\log \mathscr{D}_N$ and $\mathscr{J}_N$ in (8) is not controlled. Nor does it compare physical correlations whose numerators change with the row. The empirical W_1 form is simply a convenient exact description of the proved average displacement.

6 A finite integration certificate

The exact trace avoids a near-singular inverse but still has unbounded integration domains. A finite calculation requires a separate tail budget. The following is a constructive conditional numerical interface. Set

$$\kappa_{0,>R} = \sum_{\pm} \int_{|r|>R} |W_{\pm}(r)|\, dr, \qquad X = L + R/2 + H, \quad R, H \ge 0,$$

and let $\mathscr{J}_N^{R,H}$ denote (11) restricted to $|r| \le R,\ |x| \le X$.

Proposition 6.1 (Two truncation errors)**.** *Uniformly in* (5)*,*

$$|\mathscr{J}_N - \mathscr{J}_N^{R,H}| \le (4L + 2D_h)\kappa_{0,>R} + \frac{8}{\gamma} e^{-\gamma H}\kappa_0. \tag{20}$$

If a validated finite quadrature returns a complex value q with error at most $\varepsilon_{\mathrm{quad}}$ for $\mathscr{J}_N^{R,H}$, then $\log \mathscr{D}_N \ge \Re q - \varepsilon_{\mathrm{quad}}$ minus the right side of (20)*. Node rounding can be included by subtracting the bound* (18)*.*

Proof. Equation (15) gives $\|1 - B_a(\cdot \pm ih)\|_1 \le 2L + D_h$. Thus the absolute x integral in (11) is at most $4L + 2D_h$, uniformly in r. This controls the discarded r tail. For $|r| \le R$, $|x| > X$, each shifted real part has absolute value at least $L + H + |x| - X$, with the same sign as x. Using $|1 - AB| \le |1 - A| + |1 - B|$ and (14), the two x tails have integral at most

$$\int_{|x|>X} 4N e^{-\gamma(|x| - R/2)}\, dx = \frac{8}{\gamma} e^{-\gamma H}.$$

This proves the truncation formula. Finally $\mathscr{J}_N$ is real by its probability interpretation and continuous extension; the absolute error and Jensen give the stated one-sided certificate. □

No interval quadrature of g or numerical value of C_h is claimed executed here. Such a calculation must include Fourier inversion and quadrature errors. Taking a finite matrix or sampling a few heights alone does not supply those errors.

7 Unequal arcs and the global dependency map

7.1 What the local gain transfers

Let $\mathcal{P}_N = (\gamma/2)^N \prod_{i<j} H_*(a_i - a_j)^2$. Whenever a physical nonnegative observable has a valid source representation with absolute normalized numerator at most $\mathcal{U}_N$, division by (17) gives

$$0 \le \text{observable} \le e^{C_h} N^{-5/24} \mathcal{U}_N. \tag{21}$$

This removes the denominator's $N^{o(1)}$ loss only. In the marked-pair application $N \asymp b$; all the other $o(\log b)$ errors in (1) remain. For the one-arc integral, the source supplies only $\mathcal{U}_N \le e^{CL_Y}$ with an unspecified constant. Formula (21) does not give a new explicit polynomial exponent, because that numerator constant has not been made explicit. It is valid at arbitrarily unequal multiplicities; the numerator hypotheses, not the denominator, still determine which wall regimes are covered.

7.2 The source's physical and analytic route

The following map covers every scientific module, but reading is not independent proof. Source line locators refer to the pinned files.

Module	Mechanism and retained obligation
`01_reference`	Finite matching diagrams, disk projection of half-cylinder states, braid/fusion identities, centered vacuum polynomial and bilinear spectral star. Absolute half-state summability and the companion electric model remain imported.
`02_finite`	The marked spin sum has twist $k^{-2\sum_X h_i}$, $k = e^{3i\beta}$. Two oriented arcs join into one polygon. The chord-angle relation cancels the twist, while unmarked loops have zero orientation sum. Biorthogonal covectors and degree exhaustion produce the finite gas identity; residue strings have lengths $0, 1, 2$, with factorials and double-particle multiplicity retained. These diagram and residue calculations are read, not comprehensively independently replayed.
`03_denominator`	Positive exponential-determinant measure, Jensen and exact Cauchy trace. The uniform bounded remainder and displacement estimate are independently proved in Sections 4–6.
`04_mean`	A curved X contour carries a positive density, not an arbitrary complex contour measure. Fourier mean cancellation, branches, balayage and a multi-label real coercivity estimate are essential. The finite symbol positivity calculation and all high-frequency uniformity are retained source inputs.
`05_analysis`	Compensated empirical energies, integer Legendre reference dimensions, common/shoulder trial fields, trace-class fluctuation determinant, rare-center dyadic comparison and actual particle transport. Small singular values require a gap and a counting bound, not finite-mode agreement alone. Source companion source/quadratic/core lemmas remain imported.

Module	Mechanism and retained obligation
`06_one_arc`	Two probe contours, residues with at most two bound particles, positive half-mass neutralization and real absolute integration. Counts use L_Y but each main label starts at its own wall. No complex particle displacement is used in this coarse bound.

Several distinctions are indispensable. The star on a source spin state is the spectral involution $u \mapsto 3\lambda - u$, not a Hilbert-space adjoint. Taking absolute values of a signed gas prematurely destroys its equality with a positive polygon mass. For the one-arc contraction the source first identifies a constant unit phase times the unsigned arc weight. It then takes absolute values and may discard the attenuation factor in $[0, 1]$; discarding it before that step would assert a false signed identity.

The one-arc gas has $n - 1$ elementary particles while its background has mass $n = s + b$. The two compensators $\chi_t(u) = \frac{1}{2}\psi(u - \Re y_t)$ supply exactly the missing mass. They are positive, so the weighted cumulative field is neutral and the source's real coercivity applies. A bound particle's actual line is compared with the standard far line only after smoothing; an unsmoothed fictitious collision cannot be inserted in that estimate. Choosing one or two attached particles cancels the factors r or $r(r-1)$ against $r!$, leaving the factorials for the remaining free variables. None of these facts is implied by Cauchy interpolation.

The sharp magnetic estimate also keeps its two reference partition masses J_{N_g}, with $\log J_{N_g} = -\frac{1}{4}\log(N_g + 1) + O(1)$. The source's exponent assembly is

$$-2L_X + \frac{3\gamma}{8}L_X + \frac{\gamma}{16}(L_Y - L_X) + \frac{22}{25}(L_Y - L_X) - \frac{5\gamma}{24}L_Y + o(L_Y).$$

With $\gamma = 4/3$, the coefficient of $\log s$ is $-4/3$, and that of $\log(b/s)$ is $617/1200$. Our theorem replaces only the denominator's error by $O(1)$. It neither removes the rare-minority assumption nor controls the remaining projection, smoothing and transport errors.

Finally, the planar polygon sum uses a positive fraction of ordered marks satisfying a geometric angle and separation condition. It does not apply (1) to every pair. The short-polygon part imports a planar diameter tail from the companion manuscript. The selected long pairs have s at least a fixed power of the diameter, b comparable to it and an injective cylinder projection. This geometric transfer is separate from the denominator argument.

8 Original Ram comparison and current literature

Fresh readings of three original private reports support a restrained comparison, not a claim that they prove the physical model. TR117 [5], lines 226–339, proves the exact source/displacement change of variables and warns that localization does not invert the operator. Lines 499–632 prove absolute weighted remainder bounds and explicitly distinguish L^1 kernel mass from squared energy. In the present report the weight $|W_{\pm}(r)|(1 + |r|)$, rather than a formal finite jet, controls the discarded transition error. Thus an exact localized identity and an absolute tail budget are used together. No Taylor truncation at a complex contour node is needed.

TR018 [6], lines 637–728, retains cycle orientation, branch amplitudes and total variation when forming complex moments. Its common-domain proof at lines 1154–1243 and convolution proof at 1245–1293 justify parameter integration only under domination and specified multiplier conventions. Here those requirements are met by the S strip, the fixed h, and $\kappa_0, \kappa_1 < \infty$. There is no fractional derivative or continuation in the integer number of particles.

TR250 [7], lines 116–194 and 454–520, distinguishes unisolvent finite coordinates from the physical space and measure. Its positive polynomial Gram proof and hypercircle proof are useful contrasts: the present Cauchy matrix is bilinear and becomes singular at collisions. We do not import real-energy positivity or a flux certificate into it. The stable object is instead the cleared trace integral. The classical content of all these private passages is acknowledged; their use is not itself scored as a new theorem.

The public literature was refreshed on October 9, 2026 using only public titles and URLs. The Annals page for Duminil-Copin and Smirnov [3] confirms the honeycomb connective constant; this does not by itself establish the present marked correlation exponents. Glazman–Manolescu [4], arXiv version 3 of June 12, 2019 and the 2020 journal record, concerns Yang–Baxter weighted half-plane two-point functions and fugacity. Its boundary fugacity must not be confused with the vertex activity ρ_{v}. Ishikawa–Okada–Tagawa–Zeng [2], version 2 of September 4, 2005, records classical Cauchy/Pfaffian generalizations. These abstract and publication/version checks are not fresh full-proof audits. The current publisher README for [1] confirms its title/date, not its validity. No comprehensive priority search or independent global SAW result is claimed.

9 Advantages, costs and further work

The improvement removes a subpower denominator loss using precisely the source's existing interpolation representation. It handles all even sizes, all admissible imaginary node lists and collisions with the same constants. The average-displacement estimate permits controlled rounding of a height list without a bound for $\|K[a]^{-1}\|$. The truncation theorem states the missing analytic budgets of a finite evaluation explicitly.

The price is a one-sided certificate and contour-dependent constants. As h approaches the upper endpoint, c_0 tends to zero; as it approaches the lower endpoint the symbol decay margin tends to zero. There is no assertion of constants uniform at either endpoint. The W_1 estimate applies at fixed N, and the global Pfaffian-to-physical relation still needs the source's algebra. Computing g with rigorous errors may cost more than an ordinary floating matrix calculation; no runtime advantage was measured.

A useful next step is a validated evaluation of κ_0, κ_1 for a fixed interior choice, for example $h = 3\beta/4$, including Fourier and quadrature tails. A second is to control the Jensen excess in (8); only then could one seek full-logarithm stability or a two-sided prefactor. A third is to quantify the remaining numerator remainders without altering the prescribed ordered limits. None of these tasks is solved by the present finite fixtures or by the source's interpolation degree bounds.

10 Verification and contribution boundary

All twelve cached source texts, including the seven scientific sections, main file, bibliography, two figures and README, were displayed literally in bounded untruncated intervals. The ledger records their pre-read and post-read hashes. The original private read intervals are listed separately with their unread complements; no whole-corpus reading is asserted. Imported companion proofs, the external reference proofs and the source's full arithmetic symbol-positivity calculation were not independently audited here.

The owned exact checker exercises rational Cauchy bordered identities, confluent product expressions, the shifted-window union formula, product telescoping, node-factor derivative algebra and the final rational exponent assembly. The checker passed 5,569 parameter cases and 17,764 exact assertions; its group counts are reported separately in the owned receipt. These finite checks

supplement, not replace, the analytic proofs or the physical spin and contour arguments. Parent tests are excluded from the owned totals.

No source theorem was corrected in this report. The source denominator proof was quantitatively sharpened, and a precise lower-certificate stability statement was added. The half-state/vacuum-polynomial normalization distinction is explicit. C3 is assigned to Theorems 4.3 and 5.1 together with Proposition 6.1, not to the classical finite identities, public metadata checks or applicability L3.

References

[1] OpenAI. *Marked polygon correlations and one-arc bounds*. September 26, 2026. Cached source at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Pinned public source directory. All local scientific sections read; global results remain attributed.

[2] M. Ishikawa, S. Okada, H. Tagawa and J. Zeng. *Generalizations of Cauchy's determinant and Schur's Pfaffian*. Advances in Applied Mathematics 36 (2006), 251–287. Primary preprint and version record.

[3] H. Duminil-Copin and S. Smirnov. *The connective constant of the honeycomb lattice equals* $\sqrt{2+\sqrt{2}}$. Annals of Mathematics 175 (2012), 1653–1665. Journal abstract and publication record.

[4] A. Glazman and I. Manolescu. *Self-avoiding walk on* $\mathbb{Z}^2$ *with Yang–Baxter weights: universality of critical fugacity and 2-point function*. Annales de l'Institut Henri Poincare, Probabilites et Statistiques 56 (2020), 2281–2300. Primary preprint, abstract and version record.

[5] M. Subbarao. *A Ram Framework for Solving Integral and Integro-Differential Equations*. Private original `TR-rev-1-old-tr117.tex`, in the same-named folder under `RamFoundationTRs`. Fresh proof intervals: 226–339, 499–632; header 1–65.

[6] M. Subbarao. *Complex-Order N-Dimensional Complex Ram Transforms*. Private original `TR-rev-1-old-tr018.tex`, in the same-named folder under `RamFoundationTRs`. Fresh intervals: 1–65, 637–728, 1144–1243, 1245–1293.

[7] M. Subbarao. *Compatible Variational Ram Panels*. Private original `TR-rev-1-old-tr250.tex`, in the same-named folder under `RamFoundationTRs`. Fresh intervals: 1–65, 116–194, 454–520. The exact absolute paths and hashes are retained only in local evidence.